From "Truth Topology" to Readout Bottlenecks: Negative Replication on GSM8K and a Positive Representation–Decoder Dissociation in Procedural Micro-Worlds

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Abstract

This paper reports a multi-stage study of hidden-state geometry for correctness prediction in language-model reasoning. The original hypothesis was ambitious: perhaps global topological summaries of hidden-state traces could provide a robust signal of correctness or "truth." That hypothesis did not survive stronger checks. On GSM8K-style reasoning traces [4], dynamic and windowed H_0 features were weak, often degenerate, and did not outperform simple controls. A separate fixed-decoding GSM8K branch yielded a stronger operational result: non-convergence, as measured by token-cap termination, was the dominant predictor of error.

The main positive result emerged only after changing the task. We introduced a procedural micro-world benchmark with nonce vocabularies, held-out templates, and exact tri-label semantics over `True`/`False`/`Unknown`. In that setting, a consistent dissociation appeared across Qwen and Gemma families [7, 5]: decoder behavior systematically under-expressed `Unknown`, yet verdict-region hidden-state probes recovered substantial `Unknown` signal on held-out worlds. This dissociation survived prompt-path controls, constrained label decoding, base-vs-instruct checks, verdict-step logit analysis, and layer sweeps. The paper's final claim is therefore narrower and stronger than the original repository framing: semantic non-entailment is internally represented, but often under-realized by decoder readout.

1 Scope and Final Claim

This project began as a search for a global internal marker of correctness. The final evidence supports a different claim:

> In a held-out procedural semantic task, non-entailment (`Unknown`) is represented in hidden states but systematically under-realized by decoder outputs.

The paper therefore has three parts:

1. a negative replication record for global-topology correctness signals on GSM8K-style traces,
2. a fixed-decoding convergence analysis showing cap-linked non-convergence as a dominant error mode, and

3. a positive result on a procedural micro-world benchmark showing a replicated representation–decoder dissociation for non-entailment.

The main contribution is *not* a universal scalar "topology of truth." The stronger result is a readout-bottleneck interpretation supported by multiple controls.

2 Why the Original Idea Failed

The original working hypothesis was that correct reasoning might have a cleaner global geometric or topological signature than incorrect reasoning [3, 2]. The basic recipe was:

1. run a small model on a reasoning task,
2. collect token-level hidden states,
3. treat those hidden states as a point cloud,
4. compute topological summaries such as connected-component fragmentation (H_0) and loop structure (H_1), and
5. ask whether those summaries predict correctness.

The hypothesis was attractive, but the experiments exposed three problems.

Length confounding. Prefix-based features changed largely because more points accumulated, not because the underlying reasoning became more truthful.

Degenerate window geometry. Fixed-size window preprocessing, especially per-window standardization and whitening, often collapsed windows into nearly geometry-equivalent objects. This made several windowed H_0 summaries almost pure functions of window size.

Task contamination. On natural-language math traces, hidden states reflect too many entangled factors at once: reasoning quality, verbosity, formatting drift, stop conditions, and non-convergence. This makes a single global topology scalar too blunt to isolate correctness.

The failure of the first branch is itself important. It prevented a misleading positive claim and forced a cleaner task design.

3 Experimental Trajectory

3.1 Phase A: Global Topology on GSM8K (Negative)

Small paired runs on Qwen3.5-0.8B and Qwen3.5-2B produced weak dynamic-H_0 signals on non-capped 2B traces. Representative pilot numbers were:

- `h0_entropy_final AUC = 0.55`

- `peak_h0_entropy AUC = 0.55`
- `auc_h0_entropy_per_token AUC = 0.50`
- `max_delta_h0 AUC = 0.60`

A corrected fixed-window test did worse:

- controls-only AUC: 0.60
- corrected-topology-only AUC: 0.20
- controls plus corrected-topology AUC: 0.20

These numbers are not compatible with a strong correctness-prediction claim.

3.2 Phase B: Fixed-Decoding GSM8K (Convergence Result)

A second branch held decoding fixed for Qwen3.5-2B and shifted focus from topology to operational failure mode on a GSM8K slice [4]. The key variables were:

$$C = \mathbf{1}[\text{run terminated by } \texttt{max_new_tokens}], \qquad Y = \mathbf{1}[\text{answer is wrong}].$$

On a within-question batch of $n = 180$ samples:

- capped: 87 samples (75 wrong / 12 correct)
- EOS: 93 samples (17 wrong / 76 correct)
- AUC($C \to Y$): 0.839
- odds ratio (wrong given cap): 26.40

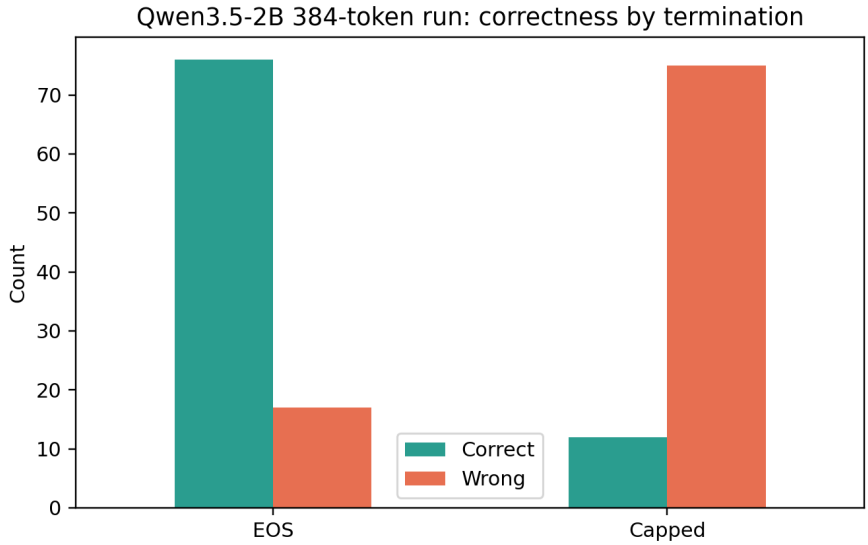

Figure 1: Fixed-config Qwen3.5-2B run: correctness split by termination mode.

The odds ratio calculation is:

$$\mathrm{OR} = \frac{\frac{75}{12}}{\frac{17}{76}} \approx 26.40.$$

A matched sensitivity rerun at 640 tokens for the originally capped cases showed that both truncation and deeper non-convergence mattered: some failures were rescued by a longer budget, but many remained wrong even after receiving more room to continue.

3.3 Phase C: Procedural Micro-World Semantics (Main Positive Result)

The project became scientifically cleaner only after replacing benchmark reasoning with a procedurally generated semantic task, in line with controlled-evaluation guidance from recent LM analysis literature [6]. Each generated world contains:

- a latent structured state,
- nonce entity names, attribute names, and relation names,
- held-out lexicons across train/dev/eval,
- held-out paraphrase templates across train/dev/eval,
- exact tri-label truth conditions over `True`, `False`, and `Unknown`.

Each world yields 72 examples: 9 propositions times 8 paraphrases. The main sweeps used 20 train worlds and 20 eval worlds.

4 Task Definition

Let a world be a finite structured state

$$w = (E, A, R),$$

where E is a set of entities, A denotes unary predicates (attributes), and R denotes binary predicates (relations). A proposition ϕ is rendered into natural language by a paraphrase template. Its semantic status is one of three classes:

$$y \in \{\texttt{True}, \texttt{False}, \texttt{Unknown}\}.$$

The labels are exact:

- `True`: the world entails ϕ,
- `False`: the world entails $\neg\phi$,
- `Unknown`: neither ϕ nor $\neg\phi$ is entailed.

This matters because `Unknown` is not a stylistic hesitation label. It is exact non-entailment under the generator.

4.1 Worked Micro-World Example

Table 1 shows one illustrative world/query slice.

Table 1: Worked micro-world example (illustrative format).

World fact set	Query statement	Gold label
`mep is falm; grel is not falm; nalo foshes sop`	`The object mep has property falm.`	True
`mep is falm; grel is not falm; nalo foshes sop`	`The object grel has property falm.`	False
`mep is falm; grel is not falm; nalo foshes sop`	`The relation fosh holds from mep to nalo.`	Unknown

5 What Is Stored for Each Example

For each example, the pipeline stores:

- the rendered prompt,
- the gold label,
- the emitted label,
- correctness and parse status,
- token IDs,
- selected hidden states,
- verdict-step logits for canonical label tokens.

The key state locations are:

- `final_prompt`
- `prompt_tail_mean`
- `verdict_token`
- `verdict_span_mean`

These are the local decision states of the model near label emission.

6 Methods

6.1 Linear Probe

Take a hidden state from one example at one chosen location:

$$h \in \mathbb{R}^d.$$

Each example provides a pair (h_i, y_i), where $y_i \in \{1, 2, 3\}$ corresponds to `True`, `False`, or `Unknown`.

A linear probe is

$$W \in \mathbb{R}^{3\times d}, \qquad b \in \mathbb{R}^3$$

with class scores

$$s = Wh + b$$

and prediction

$$\hat{y} = \arg\max_c s_c.$$

Training minimizes multiclass cross-entropy:

$$\mathcal{L}(W, b) = -\sum_i \log \frac{\exp((Wh_i + b)_{y_i})}{\sum_c \exp((Wh_i + b)_c)}.$$

This probe is intentionally weak and follows the standard linear-probe setup [1]. If it succeeds, the information is already arranged in hidden space in a directly readable linear form.

6.2 Within-World Geometry Gap

For one world and one state key, with distance $d(\cdot, \cdot)$:

$$D_{\text{same}} = \mathbb{E}[d(h_i, h_j) \mid y_i = y_j],$$

$$D_{\text{diff}} = \mathbb{E}[d(h_i, h_j) \mid y_i \neq y_j].$$

Define the world-level gap:

$$\Delta = D_{\text{diff}} - D_{\text{same}}.$$

A positive Δ means same-label states are more tightly organized than different-label states.

6.3 Verdict-Step Label-Logit Metrics

Let verdict-step logits for canonical label tokens be

$$\ell_T,\ \ell_F,\ \ell_U.$$

Then

$$p(c \mid h) = \frac{e^{\ell_c}}{e^{\ell_T} + e^{\ell_F} + e^{\ell_U}},$$

with the Unknown competitiveness margin

$$m_U = \ell_U - \max(\ell_T, \ell_F).$$

If $m_U < 0$, Unknown is under-ranked against the strongest non-Unknown candidate.

6.4 Layer Sweeps

For each layer ℓ, evaluate the same probe protocol and report

$$R_U^{(\ell)} = \text{Unknown recall of the probe at layer } \ell.$$

This reveals where non-entailment is maximally linearly recoverable in the network.

7 Main Micro-World Results

7.1 Decoder Behavior vs Hidden-State Recoverability

Table 2 summarizes the central result on held-out eval worlds.

Table 2: Decoder behavior vs verdict-token probe on held-out eval worlds ($n = 1440$ examples per model).

Model	Decoder accuracy	Decoder Unknown recall	Probe Unknown recall
Qwen3.5-2B	0.436	0.000	0.738
Qwen3.5-4B (no-think)	0.478	0.000	0.129
Gemma-3-4B-it (no-think)	0.397	0.013	0.563

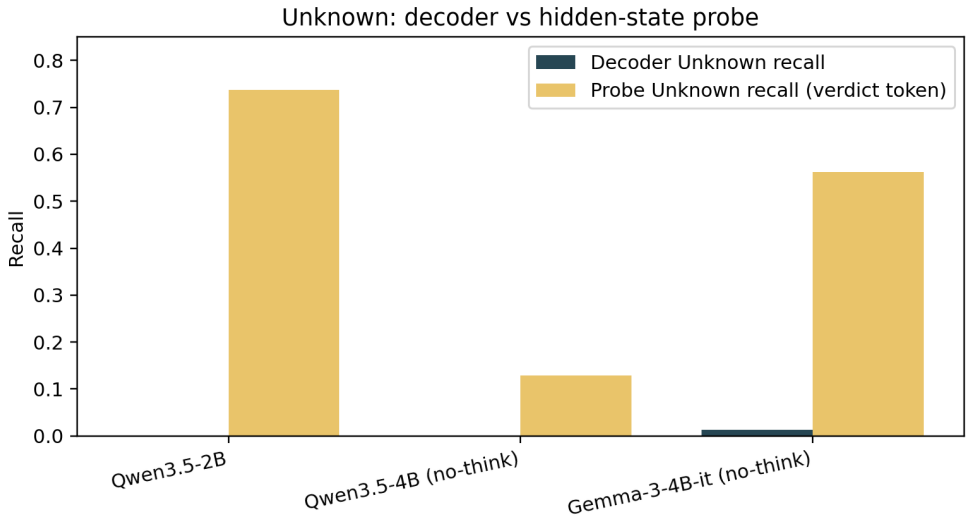

Figure 2: Unknown recall: decoder outputs vs verdict-token probe.

The key pattern is not raw accuracy. It is the gap between emitted-`Unknown` behavior and hidden-state recoverability of `Unknown`.

7.2 World-Level Geometry Consistency

World-level same-vs-different label distance gaps were positive in every evaluated world for main state keys in both Qwen3.5-2B and Gemma-3-4B-it.

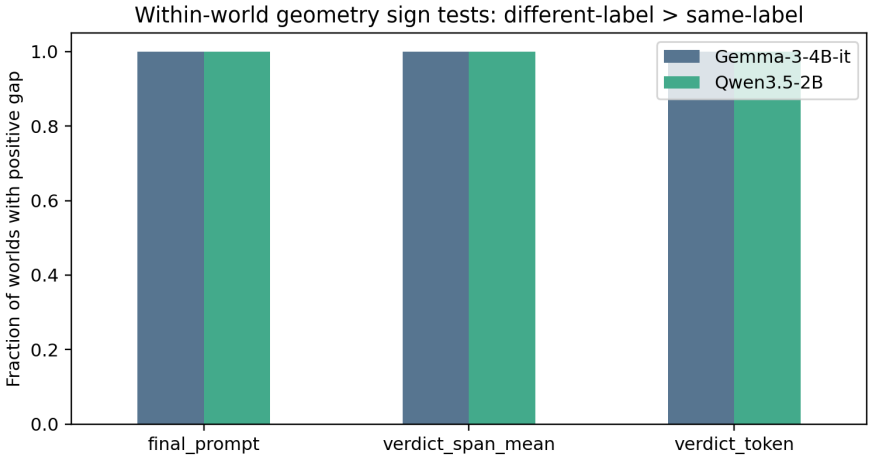

Figure 3: Fraction of worlds with positive same-vs-different label distance gap.

7.3 Cross-Family Replication

The dissociation replicates across Qwen and Gemma:

1. decoder outputs under-express `Unknown`,

2. hidden states still encode `Unknown`,
3. the gap survives held-out worlds, held-out lexicons, and held-out templates.

8 Mechanistic Controls

8.1 Constrained Decoding

To test whether free-form decoding alone caused the issue, decoding was constrained to {`True`, `False`, `Unknown`}. This did not repair Unknown collapse in the main models.

8.2 Prompt-Path Control

Raw-prompt controls were used to rule out chat-template-only explanations. The core dissociation remained.

8.3 Base vs Instruct

Gemma base initially showed severe parse failures under an unsuitable prompt path. After repairing prompt format with a base-specific label format, the parse confound disappeared. Yet base still showed decoder Unknown collapse while probes recovered substantial Unknown signal.

9 Verdict-Step Label-Logit Analysis

For gold-`Unknown` decoder failures:

- Gemma-3-4B-it: mean $P(\texttt{Unknown}) = 0.192$, mean margin $m_U = -1.293$,
- Gemma-3-4B-pt (basefmt): mean $P(\texttt{Unknown}) = 0.177$, mean margin $m_U = -1.045$.

So Unknown is often present but not competitive enough at final label-token competition.

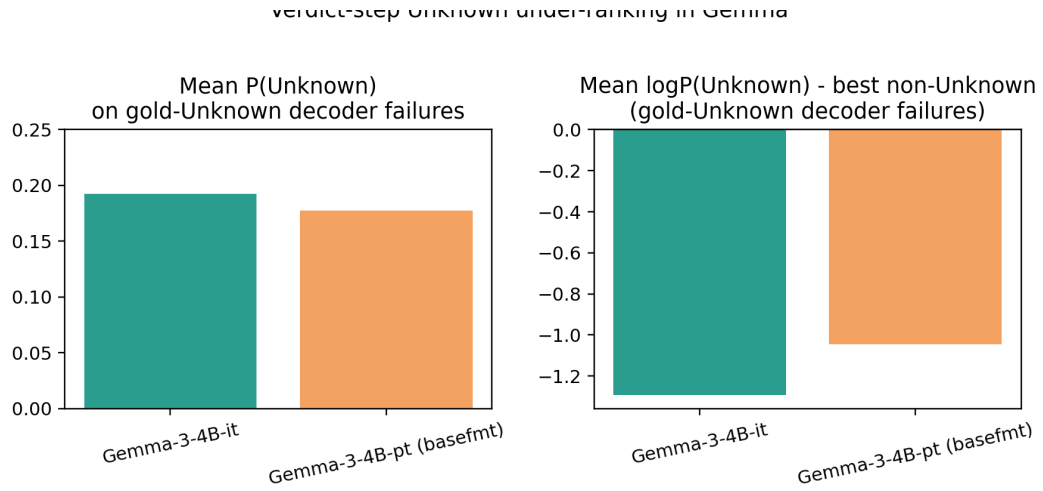

Figure 4: Unknown under-ranking at verdict step on gold-Unknown decoder failures.

10 Layer Sweeps

The best Unknown recall by state family was:

- Gemma-3-4B-it: `prompt_last` 0.779 at layer 1, `verdict_token` 0.742 at layer 10
- Gemma-3-4B-pt (basefmt): `prompt_last` 0.825 at layer 29, `verdict_token` 0.733 at layer 28

This shows strong recoverable Unknown signal exists internally in both instruct and base variants, even when emitted behavior still collapses that class.

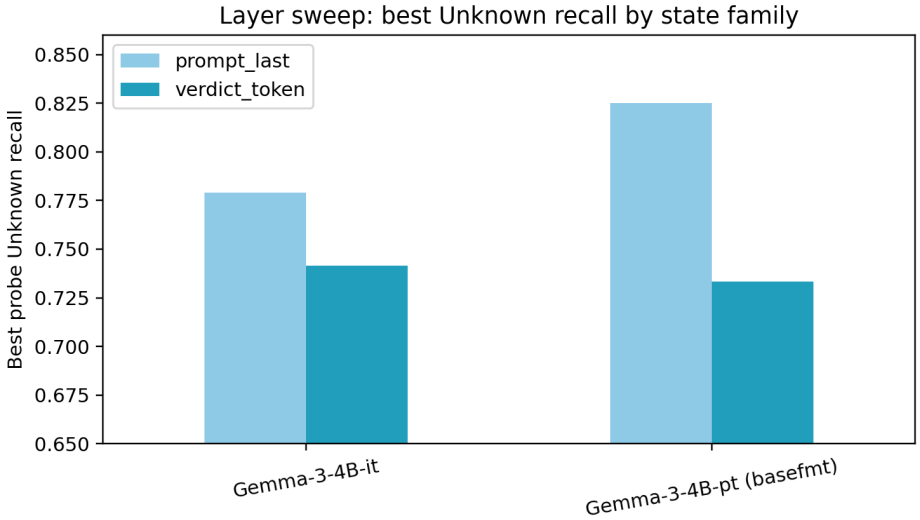

Figure 5: Best Unknown recall from layer sweeps by state family.

11 Interpretation

The combined evidence supports a readout-bottleneck interpretation:

1. semantic non-entailment is encoded internally,
2. decoder outputs under-express that class,
3. the mismatch is replicated across model families and controls,
4. verdict-step logits under-rank Unknown,
5. layer sweeps show strong internal signal upstream of failed readout.

This is stronger and narrower than the original repository framing. Results do *not* support a universal "truth topology" scalar.

12 Why This Should Survive Review

The result survives the obvious reviewer attacks:

- **Memorization objection**: nonce vocabularies, held-out lexicons, held-out templates.
- **Single-family objection**: replicated in both Qwen and Gemma.
- **Template objection**: raw-prompt controls.
- **Parse objection**: Gemma base rerun with repaired prompt format.
- **Free-decoding objection**: constrained decoding tested.
- **"No internal Unknown" objection**: probes, geometry, logits, and layer sweeps all counter it.

No single analysis carries the paper; strength comes from triangulation.

13 Practical Artifact Map

Key outputs used in the paper:

- convergence branch: `artifacts/qwen35_2b_within_question/analysis/`
- cap sensitivity: `artifacts/qwen35_2b_cap_sensitivity_640/analysis/`
- cross-model decoder comparisons: `artifacts/micro_world_v1/comparison_decoder_*.csv`
- probe comparisons: `artifacts/micro_world_v1/comparison_probe_states_*.csv`
- constrained-decoding comparisons: `artifacts/micro_world_v1/comparison_decoder_constrained_*.csv`
- verdict-step logits: `artifacts/micro_world_v1/label_logits_*/`
- layer sweeps: `artifacts/micro_world_v1/layer_sweep_*/`

14 Limitations

- The micro-world benchmark is synthetic, even though controlled and compositional.
- Model coverage is small-to-mid scale; larger-scale behavior is open.
- Probe recoverability does not prove causal use by decoder policy.
- The exact source of readout failure is not fully decomposed; it may involve calibration, label competition, or post-training behavior shaping.

15 Reproducibility

Locked dependency snapshot: `repro/requirements.lock.txt`

Figure generation:

```
python3 paper/scripts/make_figures.py
```

Compile:

```
cd paper
pdflatex -interaction=nonstopmode -halt-on-error main.tex
pdflatex -interaction=nonstopmode -halt-on-error main.tex
```

16 Conclusion

The original "truth topology" objective was too broad for the observed evidence. The final validated contribution is a replicated representation–decoder dissociation: semantic non-entailment is strongly encoded in hidden states but systematically under-surfaced at output time. The correct framing is no longer a search for a single topology-of-truth scalar, but a study of how semantic uncertainty is preserved internally and lost at decoder readout.

References

References

[1] Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes. In *ICLR Workshop Track*, 2016.

[2] Ulrich Bauer. Ripser: efficient computation of Vietoris–Rips persistence barcodes. *Journal of Applied and Computational Topology*, 5:391–423, 2021.

[3] Gunnar Carlsson. Topology and data. *Bulletin of the American Mathematical Society*, 46(2):255–308, 2009.

[4] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. arXiv:2110.14168, 2021.

[5] Google. Gemma 3 4B Instruct model card. https://huggingface.co/google/gemma-3-4b-it. Accessed 2026-04-14.

[6] Percy Liang, Rishi Bommasani, Tony Lee, et al. Holistic evaluation of language models. *Transactions on Machine Learning Research*, 2023.

[7] Qwen Team. Qwen3.5 model family card. https://huggingface.co/Qwen/Qwen3.5-4B. Accessed 2026-04-14.

A Appendix A: Full Reproduction Commands

```
# Phase A dynamic topology
python3 scripts/analyze_paired_dynamic.py \
  --artifact-08b artifacts/paired_qwen35_08b_10 \
  --artifact-2b artifacts/paired_qwen35_2b_10 \
  --out-dir artifacts/paired_dynamic_qwen35

# Phase A corrected window test
python3 scripts/analyze_2b_window_corrected.py \
  --artifact-dir artifacts/paired_qwen35_2b_10 \
  --out-dir artifacts/qwen35_2b_window_corrected

# Phase B fixed-decoding run
python3 scripts/run_multi_sample_questions.py \
  --model-id Qwen/Qwen3.5-2B \
  --question-start 0 --question-count 20 --samples-per-question 9 \
  --temperature 0.7 --top-p 0.95 --max-new-tokens 384 \
  --artifact-dir artifacts/qwen35_2b_within_question

# Phase B cap sensitivity
python3 scripts/run_cap_sensitivity.py \
  --model-id Qwen/Qwen3.5-2B \
  --source-artifact-dir artifacts/qwen35_2b_within_question \
  --out-dir artifacts/qwen35_2b_cap_sensitivity_640 \
  --temperature 0.7 --top-p 0.95 --max-new-tokens 640

# Phase C dataset
```

```
python3 scripts/generate_micro_world_dataset.py \
  --out-dir artifacts/micro_world_v1/dataset \
  --seed 1729 --train-worlds 100 --dev-worlds 25 --eval-worlds 100 \
  --props-per-world 9 --paraphrases-per-prop 8
```